

Project Report: Automated Component Transfer and Sorting Cell

1. Application Overview

- **Application Name:** Pre-Programmed Automated Component Transfer and Sorting Cell.
- **Objective:** Deploy an automated tabletop sorting system using **ROS 2 Jazzy** and the **MuJoCo physics engine** (`mujoco_ros2_control`) to execute precise, pre-programmed pick-and-place trajectories for component handling via coordinate matrices and digital triggers, bypassing real-time camera feedback.

2. Literature Review (Standard Engineering Methods)

- **Kinematics:** Standard forward kinematics compute end-effector poses from joint angles using Denavit-Hartenberg (DH) parameters or product-of-exponentials formulas. Inverse kinematics (IK) resolve joint configurations for target Cartesian coordinates using analytical solvers or numerical Jacobian-based optimization.
- **Motion Planning:** Path planning relies on sampling-based algorithms such as Rapidly-exploring Random Trees (RRT-Connect) for collision-free trajectories around static workspace obstacles.
- **Control:** Closed-loop proportional-integral-derivative (PID) or computed torque control laws regulate joint motor actuators to track planned trajectories via hardware abstractions like `ros2_control`.
- **Simulation:** High-fidelity physics-based simulation utilizes **MuJoCo** integrated directly into the ROS 2 ecosystem via `mujoco_ros2_control`, providing superior contact stability, constraint handling, and execution validation prior to physical hardware deployment.

3. Draft Project Flow (Step-by-Step)

- **Phase 1: Environment & Model Setup** — Initialize your ROS 2 Jazzy workspace, integrate the UFACTORY Lite 6 MJCF model (e.g., from MuJoCo Menagerie), and configure the `mujoco_ros2_control`

- **Phase 2: Workspace Modeling & Waypoint Programming** — Define static coordinate matrices, target bin poses, and precise joint waypoints for the component feeding tray within the MuJoCo simulation environment scene file.
- **Phase 3: Trajectory Generation** — Compute pick poses from the pre-programmed coordinate list, check reachability, and plan collision-free motion paths using OMPL planners.
- **Phase 4: Execution & Routine Control** — Execute the planned trajectory through the ROS 2 controller manager, actuate the simulated gripper mechanism using MuJoCo equality constraints/tendons, deposit the component into its designated slot, and log operational cycles.

4. Selected Robot, 6-DOF Explanation, and Justification

- **Robot Chosen:** UFACTORY Lite 6 (6-Axis Articulated Robotic Arm).
- **What does "6-DOF" (Degrees of Freedom) mean?**
 - **Degrees of Freedom** refer to the number of independent axes or joints a robot has, which dictates how it can move within a 3D space.
 - A **6-DOF** configuration means the robotic arm has six independent joint movements. This provides full control over both **position** (the X, Y, Z coordinates in 3D space) and **orientation** (Roll, Pitch, and Yaw angles). As a result, the robot's end-effector (such as a gripper) can reach a target point and approach it from virtually any arbitrary angle or direction.
- **Why you chose this robotic arm:**
 - **Compact Form Factor & Workspace Suitability:** Its desktop-friendly size and optimized reach envelope fit neatly into a controlled laboratory or desktop sorting cell without requiring industrial safety cages.
 - **Native ROS 2 & MuJoCo Compatibility:** Readily available MJCF model definitions (via MuJoCo Menagerie) paired with the modern `mujoco_ros2_control` package for Jazzy allow for rapid, high-fidelity contact simulations.
 - **Cost-Effective Dexterity:** Offering a full 6-DOF articulated structure, it provides the complex orientation handling needed for multi-axis pick-and-place tasks at a fraction of the cost of heavy industrial arms.

5. Alternative Uses for the UFACTORY Lite 6

- **Laboratory Automation & Sample Handling:** Performing repetitive tasks such as microplate moving, liquid handling, and test-tube sorting in medical or biological research labs.

- **Electronics Assembly & Soldering:** Assisting in lightweight manufacturing by holding circuit boards, plugging in cables, or guiding a soldering iron along precise trajectories.
- **Desktop Machine & 3D Printer Tending:** Automating the loading and unloading of small parts into desktop CNC machines, laser cutters, or 3D printer beds.
- **AI & Reinforcement Learning Research:** Serving as an affordable physical testbed for training machine learning algorithms in robotic manipulation, grasping optimization, and tactile feedback integration within simulation engines like MuJoCo.
- **Educational Training Platforms:** Teaching university students foundational concepts in kinematics, trajectory generation, and ROS 2 Jazzy software architecture in a safe simulation-first environment.

6. References

- **Repositories & Hardware Documentation:**
 - UFACTORY Official Developer Documentation. Available: <https://www.ufactory.cc/>
 - Google DeepMind. *MuJoCo Menagerie: UFACTORY Lite 6 Model*. Available: https://github.com/google-deepmind/mujoco_menagerie
 - ros-controls. *mujoco_ros2_control: High-fidelity simulation for ROS 2*. Available: https://github.com/ros-controls/mujoco_ros2_control